Macro

## Control Volume Balance

$$0 = \frac{dm}{dt} + \iint_{CS} (\hat{n}\cdot\underline{v})\rho\, dS$$

$$\sum_{on\,CV} \underline{f} = \frac{d(m\underline{v})_{CV}}{dt} + \iint_{CS} (\hat{n}\cdot\underline{v})\rho\,\underline{v}\, dS$$

Macro → Micro
- arbitrary CV
- Express terms as integrals
- Leibniz rule & divergence theorem

## Microscopic Balance

$$\frac{\partial\rho}{\partial t} + \underline{v}\cdot\nabla\rho = -\rho(\nabla\cdot\underline{v}) \quad \text{Mass}$$

$$\rho\left(\frac{\partial \underline{v}}{\partial t} + \underline{v}\cdot\nabla\underline{v}\right) = -\nabla p + \nabla\cdot\underline{\underline{\tau}} + \rho\underline{g} \quad \text{Cauchy Momentum}$$

$\rho$ = density
$\underline{v}$ = velocity
$p$ = pressure
$\underline{g}$ = gravity

Navier Stokes

Newtonian, Incom

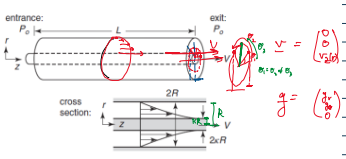
1) Choosing CV (can be arbitrary)
↓
writing out equations specific to that CV
↳ Have to rewrite & recalculate if CV is changed

→

1) Already written for an arbitrary, microscopic CV
↳ applicable to any problem ✓
↳ Just need to change boundary conditions & get velocity & pressure profile!

**PROBLEM 6.49 Flow Problem: Axial annular drag with pressure drop, wire coating.** Repeat Problem 43 with an imposed pressure gradient $-\Delta p/L$ in the flow direction. Calculate the velocity field only.

**PROBLEM: 6.43 Flow Problem: Axial annular drag, wire coating.** An incompressible, Newtonian fluid fills the annular gap between a cylindrical wire of radius $\kappa R$ and an outer shell of inner radius $R$ (Figure 6.22). The wire is pulled to the right at speed $V$. There is no pressure variation throughout the apparatus. Calculate the steady state velocity profile, the flow rate, and the force on the wire. This geometry occurs in wire-coating. Answer: [redacted]




In wire-coating, a wire is drawn through a bath. This axial drag flow is addressed in Problem 43.

1) Coord
2) $v = f(z)$
$f(r)$

$v_r = v_\theta = 0$ ; $\frac{\partial p}{\partial \theta} = 0$

r-direction

$$\rho\left(\frac{\partial v_r}{\partial t} + v_r\frac{\partial v_r}{\partial r} + \frac{v_\theta}{r}\frac{\partial v_r}{\partial \theta} - \frac{v_\theta^2}{r} + v_z\frac{\partial v_r}{\partial z}\right) = \rho g_r - \frac{\partial p}{\partial r} + \mu\left[\frac{\partial}{\partial r}\left(\frac{1}{r}\frac{\partial}{\partial r}(rv_r)\right) + \frac{1}{r^2}\frac{\partial^2 v_r}{\partial\theta^2} - \frac{2}{r^2}\frac{\partial v_\theta}{\partial\theta} + \frac{\partial^2 v_r}{\partial z^2}\right]$$

$\frac{\partial p}{\partial r} = 0$

θ-direction

$$\rho\left(\frac{\partial v_\theta}{\partial t} + v_r\frac{\partial v_\theta}{\partial r} + \frac{v_\theta}{r}\frac{\partial v_\theta}{\partial\theta} + v_z\frac{\partial v_\theta}{\partial z}\right) = \rho g_\theta - \frac{1}{r}\frac{\partial p}{\partial\theta} + \mu\left[\frac{\partial}{\partial r}\left(\frac{1}{r}\frac{\partial}{\partial r}(rv_\theta)\right) + \frac{1}{r^2}\frac{\partial^2 v_\theta}{\partial\theta^2} + \frac{2}{r^2}\frac{\partial v_r}{\partial\theta} + \frac{\partial^2 v_\theta}{\partial z^2}\right]$$

$\frac{\partial p}{\partial \theta} = 0$

z-direction

$$\rho\left(\frac{\partial v_z}{\partial t} + v_r\frac{\partial v_z}{\partial r} + \frac{v_\theta}{r}\frac{\partial v_z}{\partial\theta} + v_z\frac{\partial v_z}{\partial z}\right) = \rho g_z - \frac{\partial p}{\partial z} + \mu\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial v_z}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 v_z}{\partial\theta^2} + \frac{\partial^2 v_z}{\partial z^2}\right]$$

Cauchy - Stokes

$\frac{\partial v_z}{\partial z} = 0$ ; $\frac{\partial(\cdots)}{\partial z} = 0$

$$0 = \frac{\partial p}{\partial z} + \mu\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial v_z}{\partial r}\right)$$

$P(z)$ $\frac{\partial p}{\partial z} \approx \frac{\Delta p}{\Delta z} = \frac{\Delta p}{L}$

$\frac{\partial p}{\partial z} =$ @ $z_0 = P_0$ ; @ $z_L = P_L$ ; $\frac{P_L - P_0}{(z_L - z_0)} = \frac{\Delta p}{L}$

6.43: No pressure variation ; $\underline{v} = \begin{pmatrix} 0 \\ 0 \\ v_z \end{pmatrix}_{r,\theta,z}$ ; $\underline{g} = \begin{pmatrix} g_r \\ g_\theta \\ g_z \end{pmatrix}_{r,\theta,z}$




Mass balance / Continuity equation: Incompressible

$$\frac{\partial\rho}{\partial t} + \underline{v}\cdot\nabla\rho = -\rho(\nabla\cdot\underline{v})$$

ss.

$$\nabla\cdot\underline{v} = 0$$

$$\frac{1}{r}\frac{\partial}{\partial r}(rv_r) + \frac{1}{r}\frac{\partial v_\theta}{\partial\theta} + \frac{\partial v_z}{\partial z} = 0$$

$\frac{\partial v_z}{\partial z} = 0$

Start w/ Navier stoke:

r-direction : $0 = \rho g_r$

θ-direction : $0 = \rho g_\theta$

z-direction : $0 = \mu\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial v_z}{\partial r}\right)\right]$

Solve for velocity profile

$$\frac{\partial}{\partial r}\left(r\frac{\partial v_z}{\partial r}\right) = 0$$

$$r\frac{\partial v_z}{\partial r} = c_1$$

$$\frac{\partial v_z}{\partial r} = \frac{c_1}{r}$$

$v_z = c_1 \ln r + c_2$

Boundary conditions: @ $r = \kappa R$ ; $v_z = V$

@ $r = R$ ; $v_z = 0$

always have to be about the variable you need to solve in the PDE! (its value or its derivatives value)

@ $r = \kappa R$ $v_z = V$ → $V = c_1 \ln \kappa R + c_2$ 2eq, 2unkn

@ $r = R$ $v_z = 0$ ⇒ $0 = c_1 \ln R + c_2$

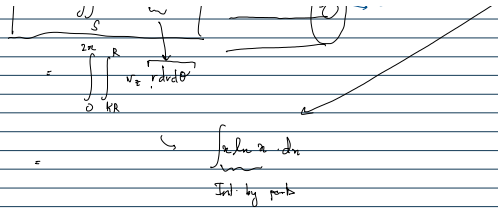
→ $c_1$ & $c_2$

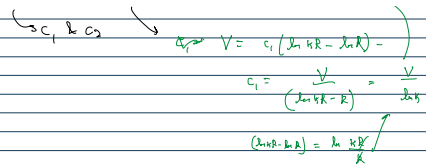
$V = c_1(\ln \kappa R - \ln R) - \cdots$

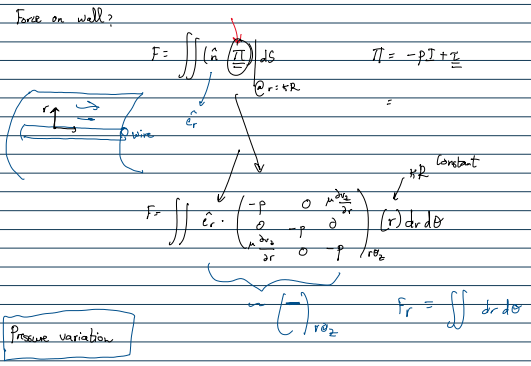
$c_1 = \frac{V}{(\ln \kappa R - \ln R)} = \frac{V}{\ln \kappa}$

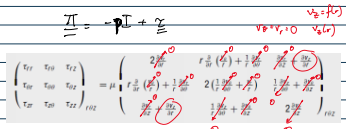
$Q = \iint_S \hat{n}\cdot\underline{v}\, dS$ ; $v_r = v_\theta = 0$, $v_z \neq 0$

$= \int_0^{2\pi}\int_\kappa^R v_z\, r\, dr\, d\theta$



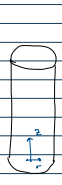


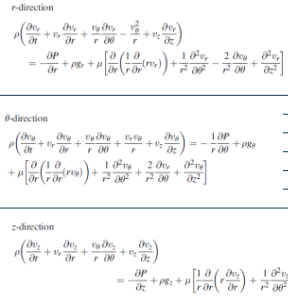




**PROBLEM: 6.52 Flow Problem: Upward pressure-driven flow in a tube, power-law non-Newtonian fluid.** Repeat problem 6.44 with a power-law, generalized Newtonian fluid with parameters $m$ and $n$.

**PROBLEM: 6.44 Flow Problem: Upward, pressure-driven flow of a Newtonian fluid in a pipe.** Calculate the velocity profile and flow rate for pressure-driven flow of an incompressible, Newtonian liquid in a vertical pipe of radius $R$. The pressure at the bottom entrance to the tube is $P_0$; at a point a distance $L$ upward the pressure is $P_L$. Assume that the flow is well developed and at steady state. Do not neglect gravity.





Power Law : Non-newtonian behavior :

Start from Cauchy balance: $\rho\left(\frac{\partial \underline{v}}{\partial t} + \underline{v}\cdot\nabla\underline{v}\right) = -\nabla p + \nabla\cdot\underline{\underline{\tau}} + \rho\underline{g}$

$\downarrow$

$\underline{\underline{\tau}} = \eta(\dot{\gamma})\,\underline{\underline{\dot{\gamma}}}$

$= \eta(\dot{\gamma})\left[\nabla\underline{v} + (\nabla\underline{v})^T\right] =$

$$\mu\begin{pmatrix} 2\frac{\partial v_r}{\partial r} & r\frac{\partial}{\partial r}\left(\frac{v_\theta}{r}\right)+\frac{1}{r}\frac{\partial v_r}{\partial \theta} & \frac{\partial v_r}{\partial z}+\frac{\partial v_z}{\partial r} \\ r\frac{\partial}{\partial r}\left(\frac{v_\theta}{r}\right)+\frac{1}{r}\frac{\partial v_r}{\partial \theta} & 2\left(\frac{1}{r}\frac{\partial v_\theta}{\partial \theta}+\frac{v_r}{r}\right) & \frac{1}{r}\frac{\partial v_z}{\partial \theta}+\frac{\partial v_\theta}{\partial z} \\ \frac{\partial v_r}{\partial z}+\frac{\partial v_z}{\partial r} & \frac{1}{r}\frac{\partial v_z}{\partial \theta}+\frac{\partial v_\theta}{\partial z} & 2\frac{\partial v_z}{\partial z} \end{pmatrix}_{r\theta z}$$

where $\dot{\gamma} = \left|\underline{\underline{\dot{\gamma}}}\right|$

$= +\sqrt{\frac{1}{2}\left(\begin{matrix}\text{Sum of Squares}\\ \text{of each term}\end{matrix}\right)}$

$\eta(\dot{\gamma}) = m\,\dot{\gamma}^{n-1}$

Momentum equation :

$$\nabla\cdot\underline{\underline{A}} = \begin{pmatrix} \frac{1}{r}\frac{\partial}{\partial r}(rA_{rr}) + \frac{1}{r}\frac{\partial A_{\theta r}}{\partial \theta} + \frac{\partial A_{zr}}{\partial z} - \frac{A_{\theta\theta}}{r} \\ \frac{1}{r}\frac{\partial}{\partial r}(r^2A_{r\theta}) + \frac{1}{r}\frac{\partial A_{\theta\theta}}{\partial \theta} + \frac{\partial A_{z\theta}}{\partial z} + \frac{A_{\theta r}-A_{r\theta}}{r} \\ \frac{1}{r}\frac{\partial}{\partial r}(rA_{rz}) + \frac{1}{r}\frac{\partial A_{\theta z}}{\partial \theta} + \frac{\partial A_{zz}}{\partial z} \end{pmatrix}_{rtz} \quad \text{B.3-10}$$

$$\tau_{rz} = \tau_{zr} = m\left(-\frac{\partial v_z}{\partial r}\right)^{n-1}\left(\frac{\partial v_z}{\partial r}\right) = \frac{(-1)}{(-1)}\,m\left(-\frac{\partial v_z}{\partial r}\right)^{n-1}\left(\frac{\partial v_z}{\partial r}\right)$$

$$= \frac{m}{(-1)}\left(-\frac{\partial v_z}{\partial r}\right)^{n-1}\left(-\frac{\partial v_z}{\partial r}\right)^{1}$$

$$\tau_{zr} = -m\left(-\frac{\partial v_z}{\partial r}\right)^{n}$$